The Hidden Variable in Embedded VLM Perception: Temporal Calibration for Real-Time Work-Zone State Estimation

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***Abstract*—Temporal smoothing constants are a hidden variable that decides comparisons between embedded perception models and between the platforms they run on. Such comparisons usually keep each system's published constants. We run a fine-tuned INT4 2B VLM, a zero-shot 4B physical-AI reasoner and a YOLO+CLIP pipeline on a Jetson AGX Orin and a DRIVE AGX Thor, on 484K annotated frames of ROADWork driving, under a latency-honest protocol in which measured latency drives frame dropping. First, temporal constants do not transfer between models: inherited from the 2B, they leave the 4B model at macro-F1 0.290; recalibrated, it reaches 0.515, and the gap holds on unseen drives. Second, the baseline needs the same treatment: recalibrated, it rises from 0.470 to 0.551 and no VLM-based system beats it, while a counted hidden-Markov filter on its score alone is statistically tied with it. Third, constants do not reliably transfer between platforms either: on the Thor they cost the hand-tuned pipeline and the 4B model 0.03–0.04 macro-F1. The recalibrated pipeline and the per-second counted filters keep their macro-F1 there, but false alarms change. The first two findings hold on drive-disjoint folds and at matched false-alarm rates, and live execution reproduces the replay on both platforms.**

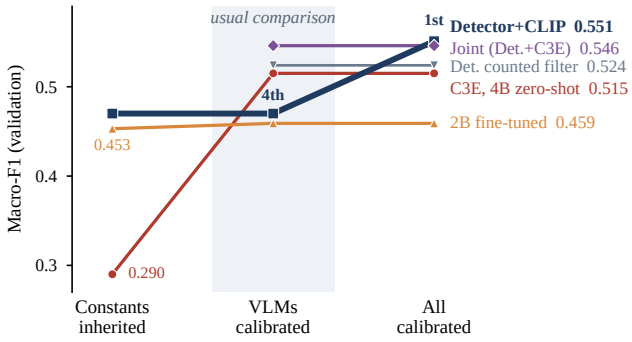


Fig. 1. The hidden variable. Validation macro-F1 (208 videos) under three comparison regimes. Left: every VLM runs the 2B's temporal constants and the detector pipeline its published ones; the zero-shot 4B reasoner (C3E) looks useless. Middle: the VLM systems are calibrated and the baseline is not, as in a typical comparison; C3E, the joint estimator and even a calibrated detector-only filter beat the published pipeline. Right: the baseline is calibrated by the same procedure and becomes the best system. Grey: the detector-only counted filter, which carries most of the joint estimator's lead.

I. Introduction

Work zones concentrate difficult driving-perception conditions: their appearance is regional and unstandardized, geometry changes daily, evidence such as cones, barrels, signs, and workers is small or occluded, and both misses and nuisance alerts are costly. They accounted for roughly 101,000 U.S. crashes and 899 fatalities in 2023 [1], yet production ADAS rarely model work zones explicitly and temporary speed limits are often absent from digital maps.

The strongest onboard approach combines a YOLO-style work-zone detector, CLIP verification [2], exponential smoothing, and hysteresis to estimate four traversal states [3], [4]. It is fast and effective after a zone is established, but reported IoU falls from 0.56/0.47 on OUTSIDE/INSIDE to 0.11/0.09 on APPROACHING/EXITING [3]. These boundaries are precisely where an advisory is most useful: an alert after the vehicle reaches the barrels has little anticipatory value.

Meanwhile, fine-tuned VLMs describe work-zone scenes well offline [5], and weight-only quantization plus embedded runtimes bring small VLMs to automotive hardware [6], [7]. Whether they should replace detector pipelines is usually judged by comparing systems as published, each with the smoothing and hysteresis constants its authors tuned. We build the strongest embedded VLM estimator we can, run it beside the detector pipeline on the same videos and protocol on two platforms, a Jetson AGX Orin and the next-generation DRIVE AGX Thor, and ask what such a comparison actually measures.

The answer is: mostly the temporal constants (Fig. 1). Against the hand-tuned detector pipeline our VLM-based systems look competitive or better, but calibrating the baseline by the same procedure reverses the ranking. A team choosing a perception stack from the usual comparison would adopt a costlier VLM stack for a gain that disappears once the baseline is calibrated.

The same variable decides comparisons between platforms. On the faster Thor, constants counted in cycles span less time and can lose macro-F1; the recalibrated pipeline and per-second counted filters keep it, but their false alarms change in ways a macro-F1 search does not see.

Our contributions are:

- **Evaluation:** a latency-honest protocol with per-model and per-platform recalibration by evidence replay, confirmed against live execution on both platforms;
- **Findings:** temporal constants transfer neither between models nor between platforms, and calibrating the baseline reverses the ranking; on unseen drives and at matched false alarms the VLM-based lead over the calibrated baseline disappears; after a platform change, the recalibrated pipeline and per-second filters keep their macro-F1, but

false alarms must be checked again;
- **Systems:** a real-time evidence-first cascade for a quantized 2B VLM, an INT4 Cosmos3-Edge reasoner on both platforms, and a counted hidden-Markov filter with per-second parameters that keeps its macro-F1 across platforms without recalibration, with an analysis of the VLMs' residual ego-relevance gap.

II. Related Work

a) Work-zone perception.: Early systems used environmental models with sensor fusion [8] or frame classification [9]. ROADWork [5] provides 4,375 videos and 9.6K annotated images from 18 U.S. cities and shows large gains from in-domain fine-tuning (+32.2 AP, +36.7 SPICE). The closest system [3] deploys a ROADWork-trained YOLO detector with CLIP verification, EMA smoothing and hysteresis, and defines the traversal-state evaluation we adopt.

b) VLMs and world models for driving.: VLMs support planning [10], [11], description [12] and reasoning [13], and Cosmos world models plan at datacenter or automotive-SoC scale [14], [15]; we estimate one safety-relevant state under an embedded latency budget. Unlike detection [16] and LLM [17] cascades, which chain models of increasing cost and escalate on confidence, our stages share one model and escalate on evidence semantics (transcribed text versus a binary answer).

c) Temporal filtering and fair comparison.: Our joint estimator is a discrete hidden Markov model [18] run as a Bayes forward filter [19]; we claim no novelty for it. That baselines deserve the proposed method's tuning budget is known [20]; we measure how much it changes conclusions for embedded temporal estimators.

III. System

A. Two embedded deployments

We deploy *two* models on the same Jetson AGX Orin 64 GB, which shares its SoC with the automotive DRIVE AGX Orin, and rerun them on a DRIVE AGX Thor (Section IX). They represent two common ways to adapt a VLM: task fine-tuning, and quantization with per-model calibration alone. Both run the same prompt channels, cascade structure, protocol and splits; only the model, its input size and its calibrated constants differ.

Fine-tuned 2B. A 2B-parameter VLM (Qwen3-VL architecture [21], initialized from Cosmos-Reason2-2B [22]) serves for perception only. It is fine-tuned through a curriculum derived from ROADWork [5], whose imagery spans 18 U.S. cities under one signing standard (MUTCD [23]). The curriculum covers work-zone VQA ($\sim$46K pairs), ego-state supervision, sign reading, a general-driving retention mix, and $\sim$10K work-zone-free BDD100K hard negatives. The last stage cut the existence gate's "yes" rate on clean BDD100K scenes from 95% to 0%. On ROADWork OUTSIDE seconds, which include visible but irrelevant work, the gate is less selective (Table VIII).

The language backbone is quantized to INT4 (W4A16 AWQ, group size 128) and the vision tower kept in FP16, both as on-device TensorRT engines [7]. Decoding (10.4 ms/token) dominates cost, followed by vision encoding (44.6 ms at 720p, $\sim$20 ms at our 480px).

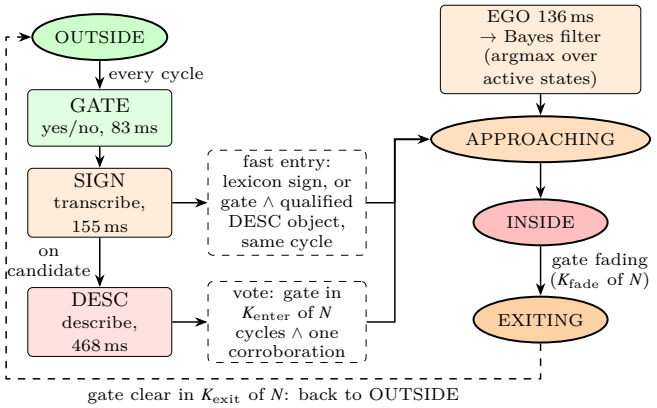


Fig. 2. Evidence-first cascade. In OUTSIDE, GATE and SIGN run every cycle and candidate evidence invokes DESC. Entry requires a sustained, corroborated vote or high-confidence same-cycle evidence. In active states, EGO drives a Bayesian filter; sustained gate clearance returns to OUTSIDE, while partial fading triggers EXITING. The constants N, K_{enter}, K_{exit}, K_{fade} are calibrated per model ($5/3/4/2$ for the 2B, $3/3/2/1$ for C3E): the structure transfers, the numbers do not. Latencies are the 2B's measured p50 at 480px.

Quantized 4B world model. Cosmos3-Edge (C3E) [14] pairs an autoregressive reasoner with a trajectory-generating diffusion transformer; we run only the reasoner, and "world model" below refers to it. It is quantized to W4A16 INT4 with an FP16 visual tower on the same Orin. Its frozen 26×46 patch grid fixes inputs at 736×416 (channel latencies in Table III). It has seen no work-zone training data. Its engines required six patches to the embedded runtime, released upon acceptance.

B. Prompt channels and their failure modes

We operate four fixed-prompt channels with fixed token budgets (the first generated token is an answer-start token).

a) Affirmation bias.: Binary prompts cannot be specialized by rephrasing. The GATE asks a neutral existence question, but path-relevance variants all answered "yes" on OUTSIDE probes and distance prompts "close" regardless of scene. Binary outputs are therefore weak evidence requiring a temporal vote, never sole triggers. A three-token answer costs $\sim$85 ms and a description $\sim$500 ms, so evidence quality, not only confidence, gates smoothing.

b) Faithful transcription.: The SIGN channel returns verbatim text such as "SHOULDER WORK" and correctly reports absence, while a binary rephrasing misses signs that transcription catches; we match transcripts against a work-zone lexicon.

c) Structured corroboration.: DESC often verbalizes disqualifying context ("barriers on right *sidewalk*", "work vehicle *parked*"), so corroboration requires a work-zone object without such a qualifier. Prompts, budgets and lexicons are released upon acceptance.

C. Evidence-first cascade

The cascade (Fig. 2) allocates compute by state. In OUTSIDE, GATE and SIGN run every cycle (Table III)

and either can invoke DESC; fast entry on a transcribed "ROAD WORK AHEAD" avoids a voting delay of up to N cycles ($\sim$1.3 s). In active states, EGO updates a Bayesian filter whose argmax is restricted to the active states, so one "outside" reply right after a sign-triggered entry cannot erase it. Only sustained gate clearance returns to OUTSIDE; partial fading triggers EXITING, which EGO never produced. The 2B's constants were fixed on calibration data. With EGO disabled, as selected for C3E, APPROACHING becomes INSIDE on the transition prior, 13 cycles after entry.

D. Joint estimator over both evidence streams

Grafting the detector and the world model onto each other's temporal machinery does not work. The ported detector counts cycles (150-cycle maximum approach, 20 minimum frames outside) tuned for its native loop. The cascade votes over a window tuned for the 2B's $\sim$260 ms cycle, while C3E answers only every $\sim$1.5 s. Either spine imports constants that are wrong for the combination.

The joint estimator therefore inherits no constant. It is a discrete hidden Markov model over the four states, run as a forward filter [18] with maximum-likelihood tables and no EMA, hysteresis threshold or cycle counter. Its observation has 108 cells. They combine the detector's frame score (six buckets), the world model's gate, sign-keyword and corroboration answers ordered by strength (six levels), and the age of that answer (three buckets). The emission is counted once per detector cycle on the calibration split with Laplace smoothing. It is counted jointly rather than factorized, because the interaction is the signal. The world model's APPROACHING is reliable when the detector corroborates it and mostly a false alarm when it does not (73% of those frames are OUTSIDE). The transition is a per-second rate matrix, counted as ground-truth transitions over dwell time and scaled by each step's dt, so it does not depend on the loop rate. The log-likelihood is scaled by dt/τ ($\tau = 0.5$ s), because a world-model answer persists across $\sim$30 detector frames. Counting it once per frame drove an early version to 135 false alarms per hour. Bucket edges and τ were set on calibration data (Section VII).

The detector runs at its native rate and is never blocked; the world model is re-polled as soon as it answers, and its last answer is held with its age. We record raw evidence with measured clocks once per video and replay the filter on it, so ages and frame drops in every result are measured ones; Section VII confirms the replay against live concurrent execution.

IV. Evaluation Protocol

a) Benchmark.: We use the detector baseline's ROADWork traversal benchmark [3]: 520 videos (4.31 h of driving) from 25 drives, with 484K interval-labeled frames. Videos are 1920×1080; all run at 30 fps except seven at 120 fps. A state-stratified 60/40 split assigns all design choices to calibration and all results to validation. Calibration has 312 videos (2.59 h). Validation has 208 videos (1.73 h, 192K frames), 13 of them pure-negative. All 25 drives occur in both partitions, so Section VIII repeats every replay-fitted comparison with drive-disjoint folds. ROADWork samples its annotated images from the same drives as its videos, so the 2B's fine-tuning images, like the baseline detector's training images, may come from benchmark drives; C3E is zero-shot. Every calibration effect we report compares a system with itself, perception fixed, and is not inflated by this overlap.

b) Latency-honest camera simulation.: After each measured inference cycle of duration Δ on the target hardware, the replay advances by Δ and drops intervening frames as a live camera would, so slower systems see sparser frames and pay latency inside the accuracy result. Predictions are held between cycles and scored per frame.

c) Metrics.: We follow the baseline's suite [3]. Frame accuracy and macro-F1 (the mean of per-state frame F1, $F_1 = 2\,\mathrm{IoU}/(1+\mathrm{IoU})$) score every frame; zone F1 is the frame F1 of any active state vs. OUTSIDE. We also report per-state IoU and event precision/recall, where an interval is correct if it overlaps one of the same state. Entry offsets are the Entry MAE over all states and the median INSIDE entry offset (IN entry med.). False alarms per hour count predicted active intervals that overlap no annotated one. The lead metric is the share of INSIDE onsets preceded by $\geq$2 s of active state. The replay runs each video at its native frame rate. Per-hour rates count frames at 30 fps (validation time overstated by 3%, for all systems alike). Unless stated otherwise, all intervals are drive-level 95% paired cluster-bootstrap intervals (10,000 replications that resample drives), and "significant" means that the interval excludes zero.

V. The VLM Cascade Against the Published Baseline

A. Comparison with the deployed detector system

Table I places every system on the same videos, device and protocol. Macro-F1 weights the four states equally and is strict at their boundaries. EXITING covers 5% of validation frames in segments of about 2 s (median), so a boundary off by 1 s can halve its IoU. Under this metric, the baseline's own published per-state IoUs [3] correspond to a macro-F1 of 0.43. Against the detector pipeline as published, the greedy 2B ties on accuracy and macro-F1 and matches its INSIDE recall. It trails on precision and false alarms, largely the ego-relevance errors of Section X: a competitive detector-free result, not a replacement. A causal INSIDE debounce recovers much of the precision (Table II). Its APPROACHING IoU is higher, as text and distant evidence appear before detector counts stabilize, a gap the recalibrated detector closes (Section VI).

B. Ablations and design validation

Table II toggles one component at a time from the sampled cascade (first row). No single component toggle moves macro-F1 by more than 0.06, and only binary SIGN, the qualifier filter and the unrestricted filter move it significantly. The sign channel must transcribe: an equal-cost binary question cuts accuracy by ten points and raises false alarms fivefold. Greedy decoding costs half an accuracy point but improves precision

TABLE I

ORIN AND THOR: VALIDATION, 208 VIDEOS, ONE PROTOCOL (%; TIMING IN S). ALL ROWS BUT THE LAST TWO ARE ON THE ORIN. DET.+CLIP: HAND-TUNED [3] AND RECALIBRATED (SECTION VI); DET. *cnt.*: DETECTOR SCORE IN THE COUNTED FILTER; 2B: GREEDY, NO POST-FILTER (RECALIBRATED: 0.459); JOINT: DETECTOR+C3E IN THE COUNTED FILTER. 2B VS. HAND-TUNED MACRO-F1: -0.018 $[-0.041, +0.003]$. F1@20, LEAD@20 (INSIDE ONSETS PRECEDED BY $\geq$2 S OF ACTIVE STATE): AT $\leq$20 FALSE ALARMS/H, WITH PERSISTENCE d = 5, 1.5, 0.25, 3, 3.25, 0.5 S PER COLUMN (SECTION VIII). THOR ROWS: THE ORIN CONSTANTS RUN LIVE ON THE THOR, EXCEPT TWO REPLAYS OF THOR RECORDINGS (SECTION IX): CAL., WHOSE ORIN CONSTANTS ARE ALSO THE THOR OPTIMUM, AND CNT., SCORED ON THE DET.+CLIP RECORDINGS (THERE 0.524 AND 25.3 FALSE ALARMS/H ON THE ORIN).

Metric	Det.+CLIP hand	Det.+CLIP cal.	Det. cnt.	2B	C3E	Joint
Frame acc.	62.9%	64.8%	**68.2%**	62.0%	65.5%	66.1%
Macro F1	0.470	**0.551**	0.524	0.453	0.515	0.546
Zone F1	0.847	**0.860**	0.843	0.839	0.853	0.855
IoU OUT	0.617	0.627	**0.652**	0.581	0.615	0.641
IoU APPR	0.124	**0.296**	0.219	0.152	0.278	0.246
IoU IN	0.544	0.552	0.573	0.532	0.574	**0.578**
IoU EXIT	0.106	0.153	0.124	0.062	0.071	**0.160**
IN ev. recall	95.7	77.8	88.6	**97.3**	87.0	85.9
IN ev. prec.	79.9	**92.5**	75.9	56.8	71.5	66.8
False alarms/h	30.9	**20.2**	22.5	56.2	64.6	25.9
Entry MAE	2.52	2.31	2.44	2.52	**2.15**	2.22
IN entry med.	3.47	**2.10**	2.73	2.67	2.50	2.27
F1@20	0.410	0.529	0.520	0.409	0.487	**0.537**
Lead@20	28.7	60.6	**71.9**	42.5	45.6	61.9
Thor macro F1	0.439	0.541	0.522	0.446	0.473	**0.549**
Thor FA/h	52.3	35.4	**21.4**	77.6	105.7	35.4

TABLE II

SINGLE-TOGGLE ABLATIONS OF THE 2B CASCADE (VALIDATION, 208 VIDEOS; –: NO INSIDE EVENT PREDICTED; BLANK: NOT MEASURED). LEAD: INSIDE ONSETS PRECEDED BY $\geq$2 S OF ACTIVE STATE. †: MACRO-F1 DIFFERS SIGNIFICANTLY FROM THE SAMPLED CASCADE.

Variant	Acc.	F1	IN prec.	FA/h	Lead
Cascade, sampled	62.5	0.464	54.1	60.1	79.4
greedy decoding	62.0	0.453	56.8	56.2	76.2
+ 1 s INSIDE debounce		0.478	76.0		
binary SIGN	52.5	0.407†	46.1	305.8	86.2
no fast entry	63.1	0.462	54.3	34.3	69.4
no qualifier filter	60.3	0.447†	53.9	69.7	81.2
no SIGN channel	63.2	0.456	53.4	47.8	82.5
unrestricted filter	61.4	0.445†	53.1	69.1	73.8
base checkpoint	48.8	0.289	–	29.2	–

and alarms, so we use it as the reproducible default. Fast entry buys anticipation, 10 more points of Lead, for 26 more alarms per hour. The base checkpoint collapses under the 2B's thresholds (Section VI).

C. Operational cost

The cascade keeps pace with a live camera: at p95 latency, patrol and active cycles both run at about 4 Hz (Table III). This is a pacing guarantee, not an alert deadline: at 30 m/s, the vehicle travels 7.8 m during one p95 patrol cycle. Per cycle, the VLM spends at least six times the detector's energy (Table III).

TABLE III

OPERATIONAL COST OF THE 2B CASCADE ON THE ORIN AT 480 PX (HEAVIEST PATH: 768 MS). POWER AND ENERGY ARE ABOVE IDLE. C3E CHANNEL LATENCIES AT 736×416: 162/182/488/556 MS (GATE/EGO/DESC/SIGN).

Latency (ms)	p50	p95
GATE / EGO	83 / 136	90 / 149
SIGN / DESC	155 / 468	169 / 508
Patrol / active cycle		259 / 239 (3.9 / 4.2 Hz)
	VLM	Detector
Power (W)	28.6	22.4
Energy per cycle (J)	3–26	0.5 (24 ms cycle)

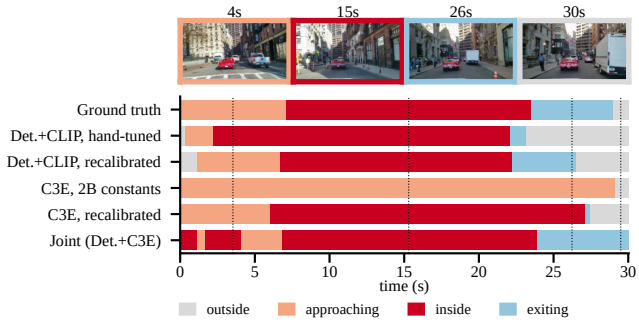


Fig. 3. Ground truth and five systems on one validation video, chosen to illustrate the effect; thumbnails mark the middle of each annotated state. With the 2B's constants, C3E never leaves APPROACHING; recalibrated, it tracks the zone. The hand-tuned detector pipeline enters INSIDE 5 s early and cuts EXITING short; recalibrated, it follows all four states.

VI. CALIBRATING EVERY SYSTEM

Section V ran each VLM with thresholds fitted to the 2B and the detector pipeline with its published constants. All systems find the work zone about equally well: zone F1 lies between 0.84 and 0.86 for every system in Table I. They differ in when they switch states, which is what the temporal constants decide. Here every system is recalibrated by the same replay procedure, and the ranking reverses (Fig. 1).

a) Temporal hyper-parameters do not transfer.: With the 2B's constants, the zero-shot C3E almost never reaches INSIDE (IoU 0.007, recall 3.8%) and looks useless. Recalibrating the same four integers and two switches lifts its INSIDE IoU to 0.574, with model, prompts and videos unchanged (Fig. 3). The optimum is shorter and less hysteretic than the 2B's and disables EGO (Section X; search space in Table IV). Because each channel is recorded once and the state machine replayed, a sweep costs CPU-seconds instead of GPU-hours.

The same search moves the fine-tuned 2B by only +0.006: it rescues a model whose constants were wrong rather than rewarding flexibility.

b) The baseline too.: The detector pipeline deserves the same treatment. We record every component on every cycle of all 520 videos (detector counts before and after CLIP verification, global CLIP and colour-context scores) and replay the complete pipeline; its hand-tuned configuration reproduces the live run (Table IV). Searching its thresholds, EMA rate

TABLE IV

RECALIBRATION BY ONE REPLAY PROCEDURE: VALIDATION MACRO-F1 OF A AND B AND THEIR PAIRED DIFFERENCE (†: INTERVAL EXCLUDES ZERO). *EACH RECALIBRATED ON ITS OWN EVIDENCE, SCORED ON HELD-OUT DRIVES OF THE SEARCH VIDEOS. REPLAY OF THE HAND-TUNED PIPELINE: 0.473 (LIVE: 0.470). VLM SEARCH: $N \in \{3, 5, 7, 9\}$ (LATER ALSO $\{1, 2, 4\}$, SAME OPTIMUM), $K_* \in 1..N$, EGO/SIGN SWITCHES; 4,896 CONFIGURATIONS ON THE 100 CALIBRATION VIDEOS WITH RECORDED VLM EVIDENCE; C3E OPTIMUM N=3, K=3/2/1. DETECTOR SEARCH: THREE GRIDS OF 4,800 CONFIGURATIONS ON ALL 312 CALIBRATION VIDEOS.

A → B	A	B	ΔF1
Det.+CLIP: hand → recal.	0.470	0.551	$+0.081^\dagger$ $[+0.043, +0.111]$
same, VLM budget	0.470	0.522	$+0.052^\dagger$ $[+0.038, +0.070]$
Det. score: hand → recal.	0.435	0.530	
2B: original → recal.	0.453	0.459	$+0.006$
Base → fine-tuned*	0.377	0.493	$+0.117^\dagger$ $[+0.039, +0.185]$
Det.+CLIP → C3E, recal.	0.551	0.515	-0.036 $[-0.080, +0.005]$

and duration limits on the calibration videos raises validation macro-F1 significantly, by 0.081. With the highest INSIDE precision and the fewest false alarms (Table I), it becomes the strongest system we measure. The same search lifts the detector score alone, without CLIP, by a similar margin.

The procedure also isolates what fine-tuning buys. Recalibrated on its own evidence and scored on held-out drives of the search videos, the base checkpoint stays significantly below the fine-tuned model, by 0.117 macro-F1 (Table IV); its collapse under the fine-tuned thresholds (Table II) overstates the gap.

All choices are made on calibration data, VLM thresholds on the 100 videos with recorded VLM evidence and detector thresholds and counted tables on all 312, so the budgets favour the detector. Given the VLM's budget (one grid, the same 100 videos), the recalibrated pipeline still gains significantly over its hand-tuned version and is level with C3E (Table IV).

c) No configuration dominates.: Calibrated, the world model and the detector pipeline tie on macro-F1 (Table IV) and fail differently. C3E leads INSIDE IoU and entry MAE; the detector pipeline leads precision and nuisance, with less than a third of C3E's false alarms (Table I). Section VII combines them.

VII. COMBINING THE DETECTOR AND THE WORLD MODEL

The joint estimator significantly beats the published pipeline ($+0.076$ macro-F1) and the calibrated world model it builds on ($+0.031$). It ties the recalibrated pipeline, trading precision for recall: it finds more INSIDE events but raises more false alarms (Tables I and V).

a) Sensitivity.: Of the joint estimator's two settings, only τ matters. Coarser or finer detector buckets move macro-F1 by less than 0.01. The time constant τ instead trades accuracy for nuisance: halving it raises false alarms by about 70%, and doubling it costs macro-F1 (Table V).

b) Concurrent execution.: The replay is faithful under real GPU contention. We ran both models truly concurrently on the Orin, in wall-clock time from a decoded camera stream,

TABLE V

JOINT ESTIMATOR (DET.+C3E, COUNTED FILTER) ON VALIDATION, 208 VIDEOS. TOP: PAIRED DIFFERENCES IN MACRO-F1 AND FALSE ALARMS PER HOUR WITH DRIVE-LEVEL 95% INTERVALS (†: EXCLUDES ZERO). MIDDLE: VARIANTS OF THE DEFAULT (SIX DETECTOR BUCKETS, $\tau = 0.5$ S; TABLE I); ITS 108-CELL TABLES ARE SPARSE, WITH 20 CELLS HOLDING 97% OF THE OBSERVATIONS. BOTTOM: THE RECORDED RUNS BEHIND EVERY REPLAY VS. BOTH MODELS LIVE AND CONCURRENT ON THE ORIN (IDLE: 11.1 W).

Difference	ΔF1	ΔFA/h
Joint estimator vs.		
Det.+CLIP, hand	$+0.076^\dagger$ $[+0.038, +0.104]$	
C3E, calibrated	$+0.031^\dagger$ $[+0.006, +0.058]$	
Det.+CLIP, recal.	-0.005 $[-0.046, +0.033]$	
live vs. its replay	$+0.002$ $[-0.011, +0.020]$	$+0.6$ $[-4.7, +5.6]$
without detector	-0.086	
without world model	-0.022	25.9→22.5
Det. counted filter vs.		
Det.+CLIP, recal.	-0.026 $[-0.062, +0.009]$	$+2.2$ $[-8.1, +11.1]$
Det. score, recal.		$-11.2^\dagger$ $[-24.6, -2.1]$

Variant	F1	FA/h
3 / 10 det. buckets	0.538 / 0.545	
$\tau = 0.25$ / 1 s	0.538 / 0.504	44.7 / 22.0
5-fold CV (calib.)	0.535 (s.d. 0.019)	
Causal TCN [24]	0.550	71.9

Orin execution	Recorded	Live, both	Det. alone
Detector cycle p50/p95 (ms)	44/59	53/58 (19 Hz)	24
World-model answer every	1.54 s	1.44 s	–
F1, joint / det. cnt.	0.546 / 0.524	0.548 / 0.522	–
Power, median (W)	–	39.0 (peak 46.9)	33.5

on all validation videos. Sharing the GPU slows the detector cycle from 24 to 53 ms, similar to the recorded runs behind every result (Table V). Live, the joint estimator matches its replay within a narrow interval ($+0.002$ macro-F1, $[-0.011, +0.020]$) with no significant change in false alarms, and the detector-only filter matches its replay as closely. Adding the world model costs 5.5 W over the detector alone.

c) Where the gain comes from.: The detector carries the filter, and the world model adds early warning where the detector is weak. Without the detector, macro-F1 drops by 0.086 and EXITING almost vanishes (IoU 0.017); without the world model, macro-F1 drops by only 0.022 and false alarms fall. The detector score alone in the counted filter is the most efficient option here. Without CLIP or a threshold search, it ties the recalibrated detector+CLIP pipeline on both macro-F1 and false alarms. It also raises significantly fewer false alarms than the state machine recalibrated on the same score, 11.2 fewer per hour. The world model pays off ahead of the zone: on the 10 validation onsets preceded by a transcribed work-zone sign, the joint estimator and C3E warn $\geq$2 s ahead on all ten, the recalibrated detector+CLIP pipeline on eight.

d) A learned alternative.: A causal three-stage TCN [24] on the same evidence ties the best macro-F1 (0.550) but raises nearly three times as many false alarms as the joint estimator (Table V).

TABLE VI

DRIVE-DISJOINT 5-FOLD REPLAY, 520 VIDEOS; SYSTEMS FITTED WITHOUT THE TEST DRIVES. F1@20: MACRO-F1 AT $\leq$20 FALSE ALARMS/H. RECALIBRATION GAINS: $+0.054^{\dagger}$ $[+0.026, +0.078]$ (DET.+CLIP), $+0.089^{\dagger}$ $[+0.058, +0.114]$ (DET. SCORE); COUNTED VS. RECALIBRATED DET.+CLIP: -0.010 $[-0.036, +0.017]$ (DET.), $+0.002$ (DET.+C3E). C3E, SAME FOLDS WITHIN THE 312 CALIBRATION VIDEOS (60–68 TEST VIDEOS EACH): 0.268 WITH THE 2B'S CONSTANTS, 0.485 SEARCHED, $+0.217^{\dagger}$ $[+0.190, +0.244]$; $+12.9^{\dagger}$ $[+9.0, +18.5]$ FALSE ALARMS/H. †: EXCLUDES ZERO.

System	F1	IoU IN	IN prec.	FA/h	F1@20
Det.+CLIP, hand-tuned	0.481	0.570	79.8	26.1	0.417
Det.+CLIP, recalibrated	0.535	0.560	**90.7**	**18.7**	**0.535**
Det. score, hand-tuned	0.434	0.546	73.3	47.0	0.410
Det. score, recalibrated	0.523	0.543	86.7	43.9	0.420
Det., counted filter	0.524	0.590	73.5	22.5	0.520
Det.+C3E, counted filter	0.537	0.591	70.8	28.7	0.523

VIII. DO THE CONCLUSIONS HOLD?

a) Unseen drives.: The conclusions hold on drives a system was never fitted on. Splitting the 25 drives into five folds of about 104 videos, we refit the detector state machines (full three-grid search) and the counted filters on four folds and replay them on the fifth (Table VI). Recalibration still raises the detector+CLIP pipeline and the detector-score state machine significantly, by 0.054 and 0.089 macro-F1, and both counted filters tie the recalibrated pipeline. On the validation videos, drive-disjoint fitting moves each refitted system's macro-F1 by at most 0.010 from Table I.

The model finding holds on unseen drives too. With per-channel evidence now recorded for all 312 calibration videos, we search C3E's constants on four of the same drive folds and replay them on the fifth. Every fold selects this paper's constants, except the SIGN switch in one fold. On held-out drives they lift C3E significantly over the 2B's constants, from 0.268 to 0.485 macro-F1, at 12.9 more false alarms per hour (Table VI).

b) Matched nuisance.: The calibrated systems also lead when all systems run at the same false-alarm rate. Their native rates differ widely (Table I), so we apply one causal persistence filter to all, showing an active interval only after it lasts d seconds, until false alarms fall to 20 per hour. On unseen drives, the recalibrated pipeline, the joint estimator and the counted detector filter then keep 0.52–0.54 macro-F1, against 0.42 for the hand-tuned pipeline (last column of Tables I and VI). Matching nuisance also reorders anticipation (Lead@20 in Table I). The VLM systems and the recalibrated pipeline need a longer persistence, which delays their warnings, while the counted detector filter needs the shortest one and most often warns early.

IX. MOVING TO A NEW PLATFORM

The Thor runs every model faster than the Orin, while answers agree on 96–100% of identical frames: the change lies in timing, and replaying Thor recordings reproduces its live runs (Table VII).

TABLE VII

ORIN$\rightarrow$THOR WITH ORIN CONSTANTS (VALIDATION, 208 VIDEOS): THOR MACRO-F1, ITS CHANGE FROM THE ORIN (†: INTERVAL EXCLUDES ZERO) AND FALSE ALARMS PER HOUR, ORIN$\rightarrow$THOR. ^{r}: REPLAY OF PER-CYCLE RECORDINGS ON BOTH DEVICES (DETECTOR CYCLE MEDIAN 35 MS ON THE ORIN, 21 MS ON THE THOR); OTHER ROWS RUN LIVE. LIVE CYCLES, ORIN$\rightarrow$THOR: C3E PATROL 718$\rightarrow$240 MS, 2B PATROL 238$\rightarrow$184 MS, DETECTOR 24$\rightarrow$16 MS, JOINT LOOP 44$\rightarrow$12 MS. ΔFA/H OF THE REPLAYS: $+21.9^{\dagger}$ $[+9.3, +31.9]$ (HAND), $+15.2^{\dagger}$ $[+8.6, +21.7]$ (RECAL.), $-3.9^{\dagger}$ $[-7.1, -1.4]$ (DET. ONLY); WITHIN THE THOR, RECAL. $-$ HAND: $+0.099^{\dagger}$ $[+0.072, +0.119]$ MACRO-F1, -15.2 FA/H; DET. ONLY $-$ RECAL.: -0.019 $[-0.045, +0.007]$, $-14.1^{\dagger}$ $[-24.1, -3.6]$ FA/H. JOINT, 90% INTERVAL: $[-0.015, +0.024]$.

System	Thor F1	ΔF1	FA/h
Cycle-counted constants			
Det.+CLIP, hand	0.439	$-0.031^{\dagger}$ $[-0.037, -0.023]$	30.9$\rightarrow$52.3
Det.+CLIP, hand^{r}	0.442	$-0.031^{\dagger}$ $[-0.039, -0.022]$	28.7$\rightarrow$50.6
2B, greedy	0.446	-0.007 $[-0.021, +0.008]$	56.2$\rightarrow$77.6
C3E, calibrated	0.473	$-0.042^{\dagger}$ $[-0.055, -0.026]$	64.6$\rightarrow$105.7
Det.+CLIP, recal.^{r}	0.541	-0.010 $[-0.029, +0.012]$	20.2$\rightarrow$35.4
Per-second counted filters			
Joint (Det.+C3E)	0.549	$+0.003$ $[-0.018, +0.028]$	25.9$\rightarrow$35.4
Det. only^{r}	0.522	-0.002 $[-0.005, 0.000]$	25.3$\rightarrow$21.4

a) Carried-over cycle constants lose accuracy.: Constants counted in cycles span less time on a faster device, and the loss tracks how much each cycle shrank. C3E's patrol cycle shrinks threefold and it loses 0.042 macro-F1, the detector's shrinks by a third (-0.031), and the 2B's least (a tie). False alarms rise for all three.

b) The recalibrated pipeline transfers, its alarm rate does not.: The Orin-recalibrated pipeline keeps its macro-F1 on the Thor (a tie), and re-searching the same grids on a Thor recording of the calibration videos selects exactly the same constants. It still beats the hand-tuned pipeline significantly there, by 0.099, but its false alarms rise significantly, by 15.2 per hour, a change a macro-F1 search cannot see.

c) Per-second filters transfer.: The joint estimator's loop shrinks most, almost fourfold, yet it keeps its macro-F1, while its false alarms rise about as much as the 2B's. The detector-only filter keeps its macro-F1 (-0.002, $[-0.005, 0.000]$) with significantly fewer false alarms, and on the Thor it ties the recalibrated pipeline with 14.1 fewer per hour, a significant gap.

X. FAILURE ANALYSIS: THE EGO-RELEVANCE GAP

Most residual false positives we inspected are correct descriptions of objects irrelevant to the vehicle's path: a parked work vehicle 60 m away, sidewalk work behind barriers, detour signs for another street. The qualifier filter catches only what the model verbalizes, and binary or distance prompts do not recover the judgment.

a) Response bias.: Fine-tuning made the gate say "yes" more often while tracking the state worse; the fine-tuned model's advantage comes from its description. Recording the three checkpoints on the same videos separates how often a model says "yes" from how well it tracks the state (Table VIII). The base 2B rarely says yes outside a zone and separates INSIDE from OUTSIDE best (balanced accuracy 0.84 vs. 0.75

TABLE VIII
GATE "YES" RATE BY TRUE STATE, INSIDE-VS-OUTSIDE BALANCED ACCURACY OF THE BINARY ANSWER (BA), AND DESC CORROBORATION ON OUTSIDE; SAME 100 CALIBRATION VIDEOS, 1 HZ.

	GATE "yes" when truth is					
Checkpoint	OUT	APPR	IN	EXIT	BA	DESC
2B base	3.8%	53.1%	71.8%	9.8%	**0.84**	31.3%
2B fine-tuned	42.4%	85.3%	92.2%	51.2%	0.75	**10.7%**
C3E zero-shot	29.4%	93.3%	94.0%	34.4%	0.82	35.7%

after fine-tuning). Fine-tuning raises its OUTSIDE "yes" rate more than tenfold; since ROADWork's OUTSIDE includes visible but irrelevant work, such a "yes" is often correct perception without ego-relevance. The fine-tuned description, in contrast, corroborates on about a third as many OUTSIDE seconds as the base model's.

C3E, never fine-tuned, transcribes signs correctly yet labels most verified INSIDE frames "off-road/sidewalk" in ego-lane probes. Perceiving objects and judging whether they claim the ego path are distinct capabilities, and no prompt we tried bridged them in either model. The joint estimator tracks EXITING only through the detector, whose score decays as objects recede (Section VII). ROADWork's 129K path annotations [5] could supply trajectory-grounded ego-relevance labels.

XI. LIMITATIONS

Our claims are bounded in four ways. *(i) Data:* all footage is ROADWork daytime driving. Drive-disjoint folds cover the detector state machines, the counted filters and the C3E cascade, but not the 2B cascade, and the 2B's absolute accuracy may benefit from drive overlap with its fine-tuning images (Section IV). *(ii) Baseline:* our port scores 0.470 against a published 0.600. Our macro-F1 applied to the published per-state IoUs falls below our port (Section V-A), so the published figure differs in aggregation or split; on our metric, the port is at least on par with the original. *(iii) Search:* configurations are the best found on finite grids. With matched budgets, the baseline still gains $+0.052$ and is level with C3E. *(iv) Scope:* false alarms are measured mostly on videos containing work zones (13 of 208 are pure-negative). Replay is open-loop: we claim computational feasibility, not safety readiness, and closed-loop evaluation is future work.

XII. CONCLUSION

This study yields three insights. First, temporal constants belong to a model, not to a task: inherited across models they move INSIDE IoU by 0.567, an effect that holds on unseen drives. Second, the baseline must be calibrated like the proposed system: doing so lifts it from 0.470 to 0.551 and removes the VLM-based systems' lead, and a counted filter on its score alone is statistically tied with it on unseen drives. Third, temporal constants also belong to a platform: on the faster Thor, carried-over constants cost up to 0.04 macro-F1. The recalibrated pipeline, whose optimum is the same on both devices, and the per-second filters keep their macro-F1 there, but false alarms change unseen by a macro-F1 search.

The practical implication is direct. Teams should recalibrate every system on the target platform, or state its constants in seconds, compare at matched nuisance, and recheck false alarms after a platform change. The VLMs' remaining gap is ego-relevance, which trajectory-grounded supervision may close; night and adverse-weather evaluation is the next step. Upon acceptance we release the replay harness, prompts, metrics and baseline port.

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